

Allotrope · Foundation Model Roster

Seven reconstruction-based anomaly detectors · six thermal (1-band) · one hyperspectral (165-band)

Pr

Pratibimba

“reflection”

Thermal

1-band

1.3ms

ARCHITECTURE

CNN AE

NORM

Yes

LOSS

L2

MASKING

None

Cheap baseline · single-pixel anomalies

An

Antardhana

“concealment”

Thermal

1-band

0.2ms

ARCHITECTURE

CNN AE

NORM

Yes

LOSS

L2

MASKING

Random pixel

Learns from spatial context

Ti

Tirohita

“vanished”

Thermal

1-band

0.3ms

ARCHITECTURE

CNN AE

NORM

Yes

LOSS

L1

MASKING

Random pixel

Robust to training-set outliers

As

Asanskrita

“unrefined / raw”

Thermal

1-band

0.2ms

ARCHITECTURE

CNN AE · 3-ch

NORM

No (raw C)

LOSS

L1

MASKING

Channel + random

Interpretable loss in Kelvin

Dr

Drashta

“the seer”

Thermal

1-band

0.2ms

ARCHITECTURE

CNN AE · 3-ch

NORM

Yes

LOSS

L1

MASKING

Channel + random

Z-score with explicit context

Ch

Chakshu

“sight / eye”

Thermal

1-band

0.4ms

ARCHITECTURE

SegFormer MAE

NORM

Yes

LOSS

L1

MASKING

Token MAE 50%

Long-range context · diffuse hotspots

In

Indradhanu

“rainbow”

Hyperspectral

165-band

5.1ms

ARCHITECTURE

SegFormer MAE + spectral compressor (24-d)

NORM

Per-band

LOSS

L1 + λ·SAM

MASKING

Token MAE 50%

Material-specific spectral anomalies

Roster at a glance

Six thermal · one hyperspectral

Total params

8.4M across 7 models

Thermal family

6 × 1-band · L1/L2 reconstruction

Hyperspectral

1 × 165-band · L1 + λ·SAM

Masking

MAE-style on 5 of 7

Bottleneck

Indradhanu 165 → 24 channels

Loss

Combined α=0.5 (L1 + SAM)

Card accent = model family

Plain AE

Masked AE (L2)

Masked AE (L1)

Raw C

Norm + 3

SegFormer MAE

HS MAE